Arpit Sakhreliya

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EDUCATION

University of Maryland **Aug 2024 - May 2026**
MS, Electrical and Computer Engineering *College Park, USA*
Coursework: Computer Architecture, CMOS VLSI Design, Domain-Specific Architecture Design, Computer Aided Digital Design, Embedded Systems

Gujarat Technological University **Oct 2018 - July 2022**
BE, Electrical Engineering *India*

WORK EXPERIENCE

Teaching Assistant **Jan 2025 - May 2026**
University of Maryland, Computer Organization of Embedded Systems *College Park, USA*
- Instructed RISC-V microarchitecture, pipelining, cache coherence, and HW/SW co-design concepts to 50+ students across embedded processors, FPGA and memory hierarchy topics.

Embedded Software Engineer **April 2024 - Oct 2024**
Prayosha Food Services Pvt. Ltd. *Gandhinagar, India*
- Engineered and validated automotive ECU hardware including power management circuits, CAN/LIN interfaces, sensor conditioning, and protection circuits compliant with automotive EMI/EMC requirements.
- Developed multi-layer ECU PCB layouts integrating MCUs, voltage regulators, high-current drivers, and diagnostic interfaces; verified using oscilloscopes, logic analyzers, and CAN protocol tools.

CTO & Co-founder **June 2022 - April 2024**
Tesseract Robotics *Gandhinagar, India*
- Led hardware development of autonomous mobile robot systems at a bootstrapped startup, managing a 15-person cross-functional team from schematic capture through field deployment.
- Designed IPC-compliant 6-layer mixed-signal PCBs with >95% first-pass reliability integrating STM32 MCUs, Jetson Nano compute modules, power regulators, sensor interfaces, and CAN/SPI/I2C/UART communication buses with controlled-impedance routing and EMI suppression.
- Architected AGV control system (STM32H7 + F103 over CAN) with adaptive PID, encoder/IMU sensor fusion, and 4-wheel synchronization achieving <10 ms end-to-end control latency.
- Designed 2 kW DC/BLDC motor-driver stages using MOSFET half/full-bridge topologies, gate-drive networks, Miller-effect mitigation, current sensing, and protection circuits; performed thermal derating and validated dead-time control under varying load conditions.
- Engineered multi-voltage PDN with switching regulators and LDO rails; executed fault-injection validation (over-current, brownout, short-circuit, mis wire) across voltage, current, and thermal corners, hardening protection circuits against field failures.
- Automated oscilloscope data processing and validation log analysis using Python and LLM-assisted tooling, cutting root cause identification cycle time by ~60%
- Led DFM/DFT reviews, component derating and BOM management, and coordination with contract manufacturers for production builds.

IITGN Robotics Lab, IIT Gandhinagar **Oct 2021 - April 2024**
FPGA Engineer, Junior research fellow *Gandhinagar, India*
- Implemented fixed-point DSP datapaths (Kalman filter, multi-radix FIR, radix-2 FFT, cascaded control loop) on Xilinx Nexys 4 FPGA for robotic-arm control, achieving sub-1 µs processing latency.
- Architected 30 kHz/5 kHz multi-rate control logic; achieved +0.8 ns timing slack via clock-domain partitioning, STA, and CDC analysis across asynchronous domains.
- Integrated DAC8811 SPI interface (~1 MSPS) driving a triac/Class-D MOSFET Power stage on a 4-layer PCB, achieving 0.001° resolution continuous actuator control for humanoid shoulder joint.
- Built an STM32-based IMU acquisition system at 20 kHz, using fixed-point quaternion and AHRS computation, for deterministic sensor fusion and real-time control. [link]

Electrical Team Lead **Aug 2019 - Feb 2022**
ABU Robocon (Asia-Pacific Competition) *Gandhinagar, India*
- Built rugged multi-rail power distribution boards, high-current motor-control electronics, and distributed multicontroller architectures for autonomous competition robots; reduced task completion time by 75 s.
- Directed cross-functional team of 40+ members; secured 1st place out of 44 teams at National ABU-Robocon 2021.

PROJECT EXPERIENCE

TPU-v1 Inspired AI Accelerator *(Vivado, Verilog, Xilinx Artix-7, JTAG)* **Aug 2024 - Oct 2024**
- Implemented a 2D 8×8 systolic array ALU in SystemVerilog on Artix-7 FPGA, using DSP slices and BRAM, achieving an operating frequency of ~100 MHz and ~1.2 GOPs throughput with ReLU, Sigmoid, Tanh, and Max Pooling inference support.

Women Safety Device *(NRF52832, Altium)* **Aug 2024 - Oct 2024**
- Developed low-power NRF52 wearable using frequency hopping network, IMU-triggered alerts, mobile notifications, and onboard microphone for real-time safety monitoring.

PUBLICATIONS & AWARDS

- **Arpit R. Sakhreliya**, Manoj Franklin. VHDLBench: A Dataset with Rich Contextual Relationships for Training Custom LLMs. *Accepted at the International Symposium on Quality Electronic Design (ISQED), 2026.*
- Secured $8,500 Nidhi Prayas Grant for Analog-Adaptive Embedded Drive for Actuators, IIT Gandhinagar, 2022

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Assembly, Verilog, Cuda
- **Microcontrollers:** STM32fxx/hxx, NRF52/53, ESP32, Arduino, FPGA (Xilinx, NI sbrio), Raspberry Pi, Jetson Orin
- **Tools:** STM32CubeIDE, Keil, ARM toolchain, FreeRTOS, Vivado, Altium, LTspice, MATLAB, Simulink, LabVIEW
- **Protocols:** UART, SPI, I2C, CAN 2.0B, CAN-FD, RS-485, USB, Ethernet, TCP, UDP, MQTT
- **Debug & Bring-Up:** JTAG, SWD, oscilloscope, logic analyzer, board bring-up, thermal and layout validation, fault-injection testing
- **Hardware Design:** PDN design, EMI/EMC mitigation, ADC/DAC, LDOs, switching regulators, motor drivers, DFM/DFT, rigid/flex PCB